

SSP
Computer & Communication Department
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**Modern Physics
Sheet 3
Solutions**

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1. An airplane flying upwind, downwind, and crosswind shows the main principle of the Michelson–Morley experiment. A plane capable of flying at speed c in still air is flying in a wind of speed v . Suppose the plane flies upwind a distance L and then returns downwind to its starting point. Another time, the plane flies crosswind a distance L and return. Derive an expression for the time difference between the two trips.
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Solution

Upwind and downwind trip

$$t_1 = \frac{L}{c-v} + \frac{L}{c+v} = \frac{2Lc}{c^2 - v^2} = \frac{2L}{c} \left(1 - \frac{v^2}{c^2}\right)^{-1}$$

Crosswind and return trip

$$t_2 = \frac{2L}{(c^2 - v^2)^{1/2}} = \frac{2L}{c} \left(1 - \frac{v^2}{c^2}\right)^{-1/2}$$

Time difference between the two trips

$$\Delta t = t_1 - t_2 = \frac{2L}{c} \left[\left(1 - \frac{v^2}{c^2}\right)^{-1} - \left(1 - \frac{v^2}{c^2}\right)^{-1/2} \right]$$

Because $v^2/c^2 \ll 1$, the expression can be simplified using the binomial expansion $(1 - x)^n = 1 - n x$. (In our case $x = v^2/c^2$)

$$\Delta t = \frac{2L}{c} \left[1 + \frac{v^2}{c^2} - 1 - \frac{1}{2} \frac{v^2}{c^2} \right]$$

$$\Delta t = \frac{Lv^2}{c^3}$$

2. The period of a pendulum is measured to be 3.0 s in the rest frame of the pendulum.
(a) What is the period of the pendulum when measured by an observer moving at a speed of $0.95c$ with respect to the pendulum?
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Solution

Proper time frame: frame of the pendulum (Δt_p)

Time dilated frame: observer moving relative to the pendulum (Δt)

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1}{\sqrt{1 - 0.95^2}} = 3.2$$

$$\Delta t = \gamma \Delta t_p = 3.2 \times 3.0 = 9.6 \text{ s}$$

3. With what speed will a clock have to be moving in order to run at a rate that is one-half the rate of a clock at rest?
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Solution

The moving clock runs at a rate half the clock at rest so this means that:

$$\Delta t = 2 \Delta t_p$$

Also

$$\Delta t = \gamma \Delta t_p$$

So

$$\gamma = 2$$

$$\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = 2$$

Solve for v

$$\frac{1}{1 - \frac{v^2}{c^2}} = 4$$

$$1 - \frac{v^2}{c^2} = \frac{1}{4}$$

$$\frac{v^2}{c^2} = \frac{3}{4}$$

$$v = 0.866 c$$

4. A clock on a moving spacecraft runs 1 s slower per day relative to an identical clock on Earth. What is the relative speed of the spacecraft?
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Solution

$$1 \text{ day} = 24 \times 60 \times 60 \text{ s} = 86400 \text{ s}$$

$$\Delta t = 86400 \text{ s}$$

$$\Delta t_p = 86399 \text{ s}$$

$$\Delta t = \gamma \Delta t_p$$

$$\gamma = 86400/86399$$

$$\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma$$

Solve for v

$$\frac{1}{1 - \frac{v^2}{c^2}} = \gamma^2$$

$$1 - \frac{v^2}{c^2} = \frac{1}{\gamma^2}$$

$$\frac{v^2}{c^2} = 1 - \frac{1}{\gamma^2}$$

$$v = c \sqrt{1 - \frac{1}{\gamma^2}} = 3 \times 10^8 \sqrt{1 - \frac{1}{\frac{86400^2}{86399}}} = 1.44 \times 10^6 \text{ m/s}$$

5. An astronaut at rest on Earth has a heartbeat rate of 70 beats/min. What will this rate be when she is traveling in a spaceship at $0.90c$ as measured (a) by an observer also in the ship and (b) by an observer at rest on the Earth?
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Solution

(a)

The observer on the ship will measure the same heartbeat (70 beats/min) because according to him the astronaut is at rest.

(b)

- *First event: one heartbeat*
- *Second event: the next heartbeat.*
- *The two heart beats take place at the same position in the frame of the spaceship.*
- *Therefore, the spaceship frame measures the proper time*
- *The time between two successive heartbeats in the spaceship frame is $= 60/70 = 0.857 \text{ s}$*
- $\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} = \frac{1}{\sqrt{1-0.9^2}} = 2.29$
- $\Delta t = \gamma \Delta t_p$
- $\Delta t = 2.29 \times 0.857 = 1.963 \text{ s}$
- $\text{Rate} = 1 / \text{time between two beats} = 1 / 1.963 = 0.51 \text{ beat/sec}$
 $= 31 \text{ beat/min}$

6. A physics professor on Earth gives an exam to her students who are on a spaceship traveling at speed v relative to Earth. The moment the ship passes the professor, she signals the start of the exam. If she wishes her students to have time T_0 (spaceship time) to complete the exam, show that she should wait a time (Earth time) of

$$T = T_0 \sqrt{\frac{1 - v/c}{1 + v/c}}$$

before sending a light signal telling them to stop.

Solution

- First event: start of the exam
- Second event: End of the exam
- The two events take place at the same position in the spaceship frame, so the spaceship frame measures the exam proper time (T_0)
- In earth frame:

$$T_{\text{exam}} = \gamma T_0 = \frac{T_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

- But the question asks for the time at which the professor should send the signal, so
- $T_{\text{signal}} = T_{\text{exam}} - T_{\text{light trip}}$

$$- T_{\text{light trip}} = \frac{d_{\text{light}}}{c}$$

- The light must catch the trip at the end of the exam, so the light must travel the same distance that the trips travel during the exam time so

$$- d_{\text{light}} = d_{\text{spaceship}} = v T_{\text{exam}}$$

$$- T_{\text{light trip}} = \frac{d_{\text{light}}}{c} = \frac{v}{c} T_{\text{exam}}$$

$$- T_{\text{signal}} = T_{\text{exam}} - \frac{v}{c} T_{\text{exam}} = \left(1 - \frac{v}{c}\right) T_{\text{exam}} = \left(1 - \frac{v}{c}\right) \frac{T_0}{\sqrt{1 - \frac{v^2}{c^2}}} =$$

$$\frac{\left(1 - \frac{v}{c}\right) T_0}{\sqrt{1 - \frac{v}{c}} \sqrt{1 + \frac{v}{c}}} = T_0 \sqrt{\frac{1 - \frac{v}{c}}{1 + \frac{v}{c}}}$$

7. Speedo and Goslo join a migration from Earth to Planet X. It is 20.0 ly away in a reference frame in which both planets are at rest. The twins, of the same age, depart at the same time on different spaceships. Speedo's ship travels steadily at $0.950c$, and Goslo's at $0.750c$. (a) Calculate the age difference between the twins after Goslo's spaceship reaches Planet X. (b) Which twin is the older?
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Solution

In the earth frame:

$$\Delta t_{\text{speedo's trip}} = \frac{d}{v} = \frac{20 \text{ ly}}{0.95 c} = \frac{20}{0.95} \text{ years} = 21.1 \text{ years}$$

$$\Delta t_{\text{Goslo's trip}} = \frac{d}{v} = \frac{20 \text{ ly}}{0.75 c} = \frac{20}{0.75} \text{ years} = 26.7 \text{ years}$$

But their age advance by the proper time in their frames only, so we must find the proper time of each one of them in its own frame.

Speedo's frame:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1}{\sqrt{1 - 0.95^2}} = 3.20$$

$$\Delta t_p = \frac{\Delta t}{\gamma} = \frac{21.1}{3.20} = 6.6 \text{ years}$$

Goslo's frame:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1}{\sqrt{1 - 0.75^2}} = 1.51$$

$$\Delta t_p = \frac{\Delta t}{\gamma} = \frac{26.7}{1.51} = 17.7 \text{ years}$$

So, during the trip, speedo's age increased by 6.6 years while Goslo's age increased by 17.7 years but we must note that Speedo reached planet X first so his age will continue to increase until Goslo reaches planet X.

Planet X is at rest relative to the earth, so when Speedo reaches planet X his frame is the same as the earth and his age will increase by the time difference between the two trips in the earth frame

- Age increase of Speedo after reaching planet X = $26.7 - 21.1 = 5.6$ years
- Speedo's total age increase = $6.6 + 5.6 = 12.2$ years
- Goslo's age increase = 17.7 years
- Age difference = $17.7 - 12.2 = 5.5$ years
- Goslo is older by 5.5 years

8. A spaceship is measured to be 100 m long while it is at rest with respect to an observer. If this spaceship now flies by the observer with a speed of $0.99c$, (a) what length will the observer find for the spaceship?
-

Solution

$$L_p = 100 \text{ m}$$

$$v = 0.99 c$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1}{\sqrt{1 - 0.99^2}} = 7.1$$

$$L = \frac{L_p}{\gamma} = \frac{100}{7.1} = 14.1 \text{ m}$$

9. An observer on Earth sees a spaceship at an altitude of 435 m moving downward toward the Earth at $0.97c$. What is the altitude of the spaceship as measured by an observer in the spaceship?
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Solution

To measure the altitude the ship you must have a reference object like a mountain.

The proper length of the mountain is measured by the observer on earth because the mountain is at rest relative to him.

$$L_p = 435 \text{ m}$$

$$v = 0.97 c$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1}{\sqrt{1 - 0.97^2}} = 4.11$$

$$L = \frac{L_p}{\gamma} = \frac{435}{4.11} = 106 \text{ m}$$

10. A meter stick moving in a direction parallel to its length appears to be only 75 cm long to an observer. What is the speed of the meter stick relative to the observer?

Solution

$$L_p = 100 \text{ m}$$

$$L = 75 \text{ cm} = 0.75 \text{ m}$$

$$\gamma = L_p/L = 1.33$$

$$\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = 1.33$$

Solve for v

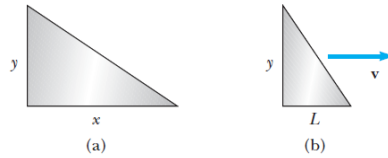
$$\frac{1}{1 - \frac{v^2}{c^2}} = 1.77$$

$$1 - \frac{v^2}{c^2} = 0.565$$

$$\frac{v^2}{c^2} = 0.435$$

$$v = 0.66 c$$

11. A spaceship in the form of a triangle flies by an observer at $0.950c$. When the ship is measured by an observer at rest with respect to the ship (Fig. a), the distances x and y are found to be 50.0 m and 25.0 m , respectively. What is the shape of the ship as seen by an observer who sees the ship in motion along the direction shown in Figure b?



Solution

The observer sees the horizontal length to be contracted but the vertical length remains as it is because the motion of the ship is in the horizontal direction only.

$$v = 0.95 c$$

$$x_p = 50\text{ m}$$

$$y_p = 25\text{ m}$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1}{\sqrt{1 - 0.95^2}} = 3.2$$

$$x = \frac{x_p}{\gamma} = \frac{50}{3.2} = 15.6\text{ m}$$

$$y = y_p = 25\text{ m}$$

12. The average lifetime of a pi meson in its own frame of reference is 2.6×10^{-8} s. If the meson moves with a speed of $0.95c$, what is (a) its mean lifetime as measured by an observer on Earth and (b) the average distance it travels before decaying, as measured by an observer on Earth?
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Solution

(a)

The proper lifetime of the pi meson is measured in its own frame of reference.

$$\Delta t_p = 2.6 \times 10^{-8} \text{ s}$$

$$v = 0.95 c$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1}{\sqrt{1 - 0.95^2}} = 3.2$$

$$\Delta t = \gamma \Delta t_p = 3.2 \times 2.6 \times 10^{-8} = 8.3 \times 10^{-8} \text{ s}$$

(b)

$$d = v \Delta t = 0.95 \times 3 \times 10^8 \times 8.3 \times 10^{-8} = 23.7 \text{ m}$$

13. A rod of length L_0 moves with a speed v along the horizontal direction. The rod makes an angle of θ_0 with respect to the x-axis. (a) Show that the length of the rod as measured by a stationary observer is given by $L = L_0 [1 - (v^2/c^2) \cos^2 \theta_0]^{1/2}$. (b) Show that the angle that the rod makes with the x-axis is given by the expression $\tan \theta = \gamma \tan \theta_0$. These results show that the rod is both contracted and rotated. (Take the lower end of the rod to be at the origin of the primed coordinate system.)

Solution

(a)

The length of the rod contracts in the direction of motion only.

$$L = \left[\left(\frac{L_0 \cos(\theta_0)}{\gamma} \right)^2 + (L_0 \sin(\theta_0))^2 \right]^{\frac{1}{2}}$$

$$L = \left[\frac{L_0^2 \cos^2(\theta_0)}{\gamma^2} + L_0^2 \sin^2(\theta_0) \right]^{\frac{1}{2}}$$

$$L = L_0 \left[\frac{\cos^2(\theta_0)}{\gamma^2} + \sin^2(\theta_0) \right]^{\frac{1}{2}}$$

$$\sin^2(\theta_0) = 1 - \cos^2(\theta_0)$$

$$L = L_0 \left[\frac{\cos^2(\theta_0)}{\gamma^2} + 1 - \cos^2(\theta_0) \right]^{\frac{1}{2}}$$

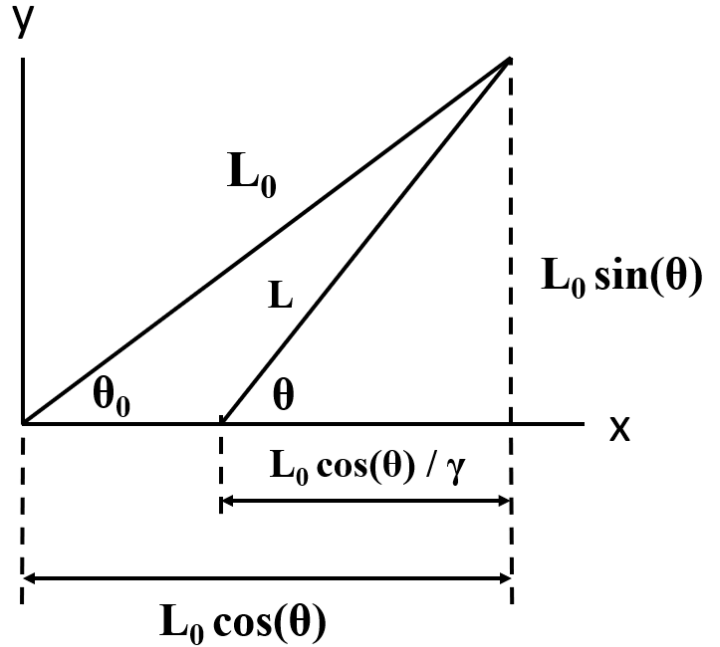
$$L = L_0 \left[1 - \left(1 - \frac{1}{\gamma^2} \right) \cos^2(\theta_0) \right]^{\frac{1}{2}}$$

$$L = L_0 \left[1 - \left(1 - \left(1 - \frac{v^2}{c^2} \right) \right) \cos^2(\theta_0) \right]^{\frac{1}{2}}$$

$$L = L_0 \left[1 - \frac{v^2}{c^2} \cos^2(\theta_0) \right]^{\frac{1}{2}}$$

(b)

$$\tan(\theta) = \frac{L_0 \sin(\theta_0)}{\frac{L_0 \cos(\theta_0)}{\gamma}} = \gamma \tan(\theta_0)$$



14. A star is receding from the earth at a speed of $5 \times 10^{-3} c$. What is the wavelength shift for the sodium D_2 line (5890 Å)?

Solution

$$\lambda_{obs} = \sqrt{\frac{1 + v/c}{1 - v/c}} \lambda_{source} = \sqrt{\frac{1 + 5 \times 10^{-3}}{1 - 5 \times 10^{-3}}} \times 5890 = 5920 \text{ Å}$$

$$\Delta\lambda = 5920 - 5890 = 30 \text{ Å}$$

- 15.(a) How fast and in what direction must galaxy A be moving if an absorption line found at 550 nm (green) for a stationary galaxy is shifted to 450 nm (blue) for A?
(b) How fast and in what direction is galaxy B moving if it shows the same line shifted to 700 nm (red)?
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Solution

(a)

The observed wavelength is smaller than the source wavelength so galaxy A is moving towards the stationary galaxy.

$$\lambda_{obs} = \sqrt{\frac{1 - v/c}{1 + v/c}} \lambda_{source}$$

Solving the equation for v/c

$$\frac{v}{c} = \frac{\lambda_{source}^2 - \lambda_{obs}^2}{\lambda_{source}^2 + \lambda_{obs}^2} = \frac{550^2 - 450^2}{550^2 + 450^2} = 0.198$$

$$v = 0.198 c$$

(b)

The observed wavelength is greater than the source wavelength so galaxy A is moving away from the stationary galaxy.

$$\lambda_{obs} = \sqrt{\frac{1 + v/c}{1 - v/c}} \lambda_{source}$$

Solving the equation for v/c

$$\frac{v}{c} = \frac{\lambda_{obs}^2 - \lambda_{source}^2}{\lambda_{obs}^2 + \lambda_{source}^2} = \frac{700^2 - 550^2}{700^2 + 550^2} = 0.237$$

$$v = 0.237 c$$